NVIDIA INCEPTION ADVANCED PROPOSAL KIT

Project Architecture: Project Koridor & The GAIINS Initiative

Submitted By: Odynex Axis Logistics LLC

Platform Focus: Federated Edge AI, Spatiotemporal Graph Neural Networks (GNNs), Computer Vision pipelines

1. Executive Summary & Core Infrastructure Framework

Odynex Axis Logistics LLC is scaling a decentralized, data-driven cyber-physical infrastructure built entirely around the intersection of high-performance accelerated computing and circular bioeconomy networks. The underlying system targets the complex macroeconomic friction of industrial and agricultural byproduct sourcing, routing, and biological conversion.

By combining real-time edge intelligence with multi-state supply chain telemetry, the infrastructure systematically converts time-sensitive agricultural waste stream liabilities into highly predictable, software-defined regional asset networks (protein feeds, organic amendments, clean energy microgrids).

2. Technical Infrastructure Matrix (GAIINS Modular Pod Architecture)

The physical implementation layer is anchored by the GAIINS Initiative—modular, containerized multi-species biological factories equipped with low-power localized edge compute engines. The platform eliminates manual inspection bottlenecks by mapping operational layers into parallel computation tasks.

Biological Subsystem	Computational AI Layer / Technical Mapping
Insect Protein Matrix (Black Soldier Fly & Mealworms)	Edge Computer Vision (CV) deployed via NVIDIA DeepStream pipelines to track real-time larval density, calculate lifecycle milestones, and automate harvesting triggers.
Precision Aquaculture (Catfish & Tilapia Cells)	IoT sensor processing and real-time inference monitoring to track metabolic stress signatures, automate precise feeding ratios, and mitigate pathogen risks.
Mycelial / Vermicompost Loops (Organic Matter Decomposition)	Predictive mass-balance modeling leveraging local nutrient-rich frass profiles and cellulose analytics to maximize real-time conversion ratios.

3. Geographically Distributed Supply Chain Graph Topology

Project Koridor implements a multi-node topological network designed to solve multi-commodity spatiotemporal routing constraints across regional corridors:

- **California Node:** Capturing orchard trimmings and rice hulls for structural insect substrate processing.
- **Texas Node:** Regulating cotton gin trash and beef cattle manure blending metrics.
- **North Dakota Node:** Sourcing fine grain dust to fuel cold-hardy mycelium inoculations.
- **Tennessee Node:** Funneling hardwood sawdust and spent brewer grains.
- **Alabama & Florida Nodes:** Diverting poultry waste, citrus pulp, and managing aquatic nutrient recovery cells.
- **SOWEGA Orchestration Hub:** Central Southwest Georgia node executing data integration, digital twin simulations, and primary infrastructure deployment.

4. NVIDIA SDK & Accelerated Architecture Alignment

The platform is engineered to align directly with the following technical SDK layers within the NVIDIA dashboard:

- **NVIDIA cuOpt:** Utilized as our core geometric routing engine to dynamically optimize cross-state material flows against transport windows, changing fleet dynamics, and biological organic degradation timelines.
- **NVIDIA JetPack & DeepStream SDKs:** Integrated directly into shipping-container micro data rigs to process multi-camera camera layouts on-site without cloud dependency, preserving rural bandwidth.
- **NVIDIA RAPIDS (cuDF/cuML):** Accelerating tabular feature engineering workloads dealing with historical agricultural yield volumes, municipal waste streams, and local climate data correlation.

5. Hard Allocated Resource Requirements

To migrate from software modeling blueprints into physical hardware-in-the-loop pilot deployments in Southwest Georgia, the following allocations are requested:

1. **Edge Prototyping Compute:** Allocation of (2) NVIDIA Jetson Orin Nano hardware development modules to serve as on-site compute blocks for real-time biological state inference within harsh agricultural conditions.
2. **Cloud Tensor Acceleration:** Provisioning of NVIDIA Cloud compute allocations (A100/H100 cloud blocks) to execute complex training iterations of spatiotemporal supply chain Graph Neural Networks (GNNs).
3. **Technical Alignment:** 1-on-1 advisory coordination with an NVIDIA systems architect specializing in accelerated operational research and routing heuristics.